The Deepfake Discount: Believing Erotic Images Are AI-Generated Reduces Affective Responses

Ana Neves[1], Marco Viola[2], Marco Sperduti[3], Guido Corradi[4], Antonio Olivera-La Rosa[4], Alessandro Ansani[4], Marco Marini[4], and Dominique Makowski[1,8]

[1]School of Psychology, University of Sussex
[2]Department of Philosophy, Communication, and Performing Arts, Roma Tre University
[3]Tor Vergata University of Rome
[4]Department of Social Psychology and Organization, Universidad Nacional de Educación a Distancia
[5]Department of Psychology and Social Sciences, Universidad Católica Luis Amigó
[6]Department of Music, Art and Culture Studies, University of Jyväskylä
[7]Institute of Cognitive Sciences and Technologies
[8]Sussex Centre for Consciousness Science, University of Sussex

The proliferation of AI-generated content raises the question of whether beliefs about a stimulus's origin alter emotional responses to it. Across two studies, we conceptually replicated and extended Marini et al. (2024) by presenting heterosexual participants (Study 1: N = 705; Study 2: N = 261) with erotic (Studies 1 and 2) and non-erotic (Study 1) images from validated databases (NAPS and NAPS ERO). Unbeknownst to participants, all stimuli were real photographs, each randomly labelled as "Photograph" or "AI-generated". Participants rated sexual arousal, enticement, and valence for each image, and later judged their agreement with the labels. Images labelled as AI-generated received lower affective ratings, but this effect was small and driven primarily by male participants. Gender differences were robust yet context-dependent: males yielded more robust differences, partially driven by a higher baseline arousal response. However, image of couples (Study 2), that females rated as more arousing, narrowed the gender gap found in Study 1. Critically, the effect of the labels interacted with participant's own beliefs: photograph-labelled images were rated more arousing than AI-labelled only when participants' subsequent belief ratings were congruent with the label, and the difference between labels reversed when opposite beliefs were reported, suggesting a complex relationship between emotion and reality beliefs. Individual-level differences in AI attitudes, pornography consumption, and sexual activity showed no reliable modulation of the labels effect. These findings suggest that the anti-AI bias is best understood as a belief-sensitive authenticity effect rather than a uniform response to AI labels, with implications for research using synthetic stimuli and for policies mandating disclosure of AI-generated content.

Keywords: AI-generated, beliefs, gender differences, affective ratings

Introduction

Generative artificial intelligence (genAI, Fisher, 2025) enables the creation of media content (e.g., images, audios, and videos) that are increasingly indistinguishable from authentic ones (Miller et al., 2023; Velásquez-Salamanca et al., 2025). The increasing availability of such content, often called deepfakes, has led to its proliferation across social networks, entertainment, immersive environments, and everyday communication (Fisher et al., 2024). Whereas deepfakes once contained clear perceptual markers of inauthenticity, such as distortions characteristic of early computer-generated contents (De Ruiter, 2021; McDonnell & Breidt, 2010), these cues are rapidly diminishing. As deepfakes reach perceptual parity with their authentic counterparts, understanding how individuals evaluate and emotionally respond to such ambiguity becomes both a theoretical and practical imperative, with implications for information integrity, social trust, and the ethical deployment of AI systems.

The prejudice against (putative) AI-generated content

Researchers have investigated the impact of authorship (human-vs. AI-generated) on perceivers' appraisal across several domains (e.g., visual art, poetry, music, faces, people). These studies typically manipulate either the *actual* origin of the stimulus (human-vs AI-generated) or the participant's *belief* about its origin, or both. Evidence has accumulated showing that participants often fail to tell apart AI-generated from human-made content at above-chance lev-

els both in artworks (Cenerini et al., 2025; Gangadharbatla, 2022; Grassini & Koivisto, 2024; Köbis & Mossink, 2021) and photorealistic images of faces created with state-of-art technology (StyleGAN3, see Stockner et al., 2026).

Interestingly, when authorship is not disclosed, AI-generated content is sometimes evaluated more positively. For instance, AI-generated poetry has been shown to receive higher appreciation ratings (e.g., beauty, lyricism, rhythm) than human-made poetry (Porter & Machery, 2024). Similarly, faces that are actually generated by AI are judged as more trustworthy than the real ones on average (Nightingale & Farid, 2022; Tucciarelli et al., 2022, study 1), possibly because generative algorithms are biased toward average, and familiar-looking faces, and familiarity breeds trustworthiness (Miller et al., 2023; but see Dunn et al., 2026).

In contrast, when stimuli are explicitly labelled as AI-generated, they are consistently evaluated more negatively than when they are presented as authentic. This "anti-AI bias" has been documented across several culturally salient domains such as visual arts (Bellaiche et al., 2023; Chiarella et al., 2022; Di Dio et al., 2023; Ragot et al., 2020), music (Ansani et al., 2025; Shank et al., 2023) and poetry (Porter & Machery, 2024). More broadly, the influence of contextual information on the evaluation of cultural products is well established and extends beyond biases specific to AI. For example, research in neuroaesthetics shows that presenting an artwork as an original rather than a (human-made) copy can alter both its perceived worth (Newman & Bloom, 2012), and associated neural responses (Huang et al., 2011). These findings suggest that evaluative processes are highly malleable to contextual cues and beliefs about the origin of the stimuli. The "anti-AI bias" makes no exception, inasmuch it seems driven primarily by cognitive evaluations tied to beliefs about authorship, as shown by the paradoxical appreciation of (unknowingly-) AI-made stimuli.

Interestingly, despite being strongly constrained by evolutionary pressures, more biological salient stimuli such as the perceived appearance of food (Califano & Spence, 2024) or human faces (Liefooghe et al., 2023), are also affected by the anti-AI bias. Califano and Spence (2024) report that, despite preferring AI-generated food images when unlabeled (especially ultra processed food), as soon as labels are introduced, participants tend to prefer authentic food pictures. Socially relevant stimuli like faces are also consistently rated as less trustworthy when presented as AI-generated as opposed to real (Liefooghe et al., 2023). Eiserbeck et al. (2023) reports that smiling faces labelled as AI-generated were judged less positively, processed more slowly, and elicited reduced early emotional EEG responses. On the contrary, subjective and EEG responses for angry faces were not modulated by the label, possibly reflecting the robustness of phylogenetically rooted threat stimuli. Similarly, Döring and Mohseni (2025) found that romantic images of heterosexual couples hanging out were rated more aesthetic, erotic, expressive, and vivid when labeled as photographs than as deepfakes. Taken together, these results suggest that beliefs about AI authenticity selectively dampen positive affective responses when dealing with biological salient stimuli. However, this conclusion should be interpreted with caution. In the studies discussed above, the positive stimuli are not typically comparable to the negative ones in terms of arousal, making it difficult to determine whether the observed effect truly reflects a selective dampening of positive affective responses.

One class of evolutionary salient positive stimuli that may offer a more appropriate comparison is erotic images, which are known to elicit strong motivational and attentional responses. Indeed, erotic images have been shown to capture attention as much as, or even more than, fear- and disgust-related stimuli (Ciesielski et al., 2010). In fact, they are a paradigmatic category of positively valence and highly arousing stimuli in many popular databases of stimuli for psychological research on emotion, be them visual (Kurdi et al., 2017; Lang et al., 1997; Wierzba et al., 2015) or audio (Holz et al., 2022; Yang et al., 2018). This makes them a particularly relevant test case for examining whether contextual information, such as an AI label, can modulate responses even to highly salient positive stimuli.

Ana Neves
Marco Viola
Marco Sperduti
Guido Corradi
Antonio Olivera-La Rosa
Alessandro Ansani
Marco Marini
Dominique Makowski

Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: Ana Neves: Data curation, Formal Analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing; Marco Viola: Conceptualization, Funding acquisition, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing; Marco Sperduti: Supervision, Writing – original draft, Writing – review & editing; Guido Corradi: Funding acquisition, Investigation, Writing – review & editing; Antonio Olivera-La Rosa: Investigation, Writing – review & editing; Alessandro Ansani: Conceptualization, Formal Analysis; Marco Marini: Conceptualization; Dominique Makowski: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing

Correspondence concerning this article should be addressed to Marco Viola, Email: marco.viola@uniroma3.it

AI and erotic stimuli

Beyond their methodological interest as biologically salient stimuli for investigating the anti-AI bias, erotic stimuli are worth investigating also for their social relevance. Indeed, genAI is often used to generate explicit visual contents (Döring et al., 2024; Lapointe et al., 2025), with an increasing number of (non-consensual) sexual deepfakes (Ajder et al., 2019; Hero, 2023). Interestingly, the widespread consumption of explicitly AI-generated sexual content would suggest that the anti-AI bias might not generalize to erotic stimuli. Or does it?

Easterbrook-Smith (2025) speculate that AI-generated pornography might still be valued less as it lacks the "aura of authenticity" surrounding human-made porn. Similarly, Viola and Voto (2023) propose that, inasmuch the spread of deepfakes cast doubts about the authenticity of all media, it will end up diminishing the allure of non-consensual sexual content. This hypothesis aligns with Tucciarelli et al. (2022)' findings, as they show how simply suspecting that some faces within a set *could* be AI-generated lowered trustworthiness score of the whole set.

Yet, to the best of our knowledge, the only empirical findings about AI-generated erotic images have been collected by Zakreski et al. (2026) and Marini et al. (2024). Zakreski et al. (2026) presented images of several kinds of female naked bodies, including real and AI-generated pictures, reporting that even if AI-generated pictures were deemed less realistic, they were nonetheless considered more attractive. However, the origin of the picture was not explicitly disclosed to participants. Moreover, since their sample was recruited from volunteers interested in sexuality and paraphilic studies in Czech Republic, they alert that it "is not representative of the general population" (p. 340).

In contrast, Marini et al. (2024) performed an online experiment presenting subjects with soft erotic images (male and female models in underwear), finding that the same images were rated higher in terms of sexual arousal when judged more realistic (in Study 1), or presented as real as opposed to AI-generated images (in Study 2). However, these studies were based on a sample from a single nation (Italy) and provide scant socio-demographic information. Moreover, they they used only mildly sexually arousing stimuli not taken from a validated dataset, and focused on a single dependent variable, namely self-reported sexual arousal. This is important as self-reported first-person, *affective* feelings ("does this arouses me?") and third-person, *semantic* judgments ("is that supposed to be arousing?") have been shown to dissociate in some cases (Itkes & Kron, 2019). Lastly, in their direct manipulation of the authentic/AI-generated labeling (Study 2) they did not provide any explicit manipulation check.

The present studies

The present studies are a conceptual replication of Marini et al. (2024), which we improved in several ways. First, we employed stimuli from a validated database, the Nencki Affective Picture System (NAPS, Marchewka et al., 2014), including explicit and more sexually arousing pictures (NAPS ERO, Wierzba et al., 2015), selected via data-driven, reproducible measures. Second, we asked participants to make different evaluations, namely first-person subjective sexual arousal and valence, as well as a third-person (more "semantic") judgment of how enticing they found the image to be.

Across both studies, we also examined whether responses to the stimuli were modulated by relevant individual differences: sociodemographic characteristics, given the well-documented sexual dimorphism in responses to erotic stimuli, with heterosexual men typically showing higher, more category-specific arousal to stimuli matching their sexual orientation than heterosexual women (Huberman et al., 2015); attitudes towards AI, given inconsistent evidence for such moderation across domains; and sexual behaviour, namely pornography consumption and recent sexual activity.

Study 1

The first study aimed at gauging whether an anti-AI bias toward erotic stimuli appears consistently in a wide and heterogeneous sample, and whether it is modulated by demographic factors, by sexual activity and pornography consumption habits, or by attitudes toward AI. Following Marini et al. (2024), we hypothesise that images presented as AI-generated will elicit lower ratings than those presented as photographs (H1).

Additionally, in line with the literature on explicit stimuli (Huberman et al., 2015), we hypothesise that erotic images relevant to participants' sexual orientation (i.e., female images for male participants and male images for female participants) would result in higher ratings than erotic but irrelevant images and non-erotic images, and that these differences would be less pronounced for women (H2).

While prior studies have examined the effect of AI labels on affective ratings, relatively little attention has been paid to whether participants actually believe those labels, and whether belief itself modulates the observed effects. If participants second-guess the authenticity claims (e.g., suspecting that "AI-generated" images are in fact photographs, or vice versa), label-based effects would be attenuated or reversed regardless of the manipulation's objective content. Indeed, Tucciarelli et al. (2022) showed that when participants were informed that some faces in a set might be AI-generated, their responses were driven not by the objective category of each face, but by their own judgment of whether that specific face was real. Accordingly, we assess the role of individual

belief in the label as an additional predictor of affective ratings (H3).

Research on general attitudes towards AI and its moderating role on ratings is mixed, with some studies finding a moderating effect (Döring & Mohseni, 2025) and others not (Ansani et al., 2025; Cloudy et al., 2022; Liefooghe et al., 2023; Marini et al., 2024). As such, we exploratory test whether attitudes towards AI moderate responses to the stimuli (H4). Lastly, based on previous findings that both pornography consumption and sexual activity shape affective and attentional responses to sexual content (Carvalho et al., 2018; Kunaharan et al., 2017; Prause et al., 2015; Walton & Bhullar, 2018; Youn, 2006), we explore whether the ratings of sexual stimuli is affected by either pornographic consumption or sexual activity (H5).

Materials and Methods

Participants

The initial sample comprised 1,067 participants recruited via multiple channels, including university research participatory schemes, social media platforms, university classrooms, and snowball sampling. This strategy yielded a heterogeneous pool of both incentivised and non-incentivised individuals, including students and members of the general population from England, France, Italy, Colombia, and Spain. Data collection occurred from the 19th of January to 1st of June 2024. At this time, publicly available image generators such as OpenAI's DALL-E 3 (Betker et al., 2023) were producing highly photorealistic outputs.

Eight participants were removed if they showed no variation in arousal ratings across trials, and four because they displayed a salient negative correlation between arousal and enticement alongside lower arousal ratings for erotic compared to neutral stimuli, suggesting a possible misunderstanding of response scales direction. In order to preserve homogeneous and well-powered samples, participants identifying with a gender other than Female or Male (N = 15) were excluded from further analysis, as were female and male participants reporting a sexual orientation other than Heterosexual (N = 335). The latter comprised 150 bisexual females, 72 bisexual males, 32 homosexual females, 38 homosexual males, and 43 participants (23 females, 20 males) who reported another orientation. The final sample consisted of 705 heterosexual participants (Mean age = 30.2 ± 11.8 years; 35.7% female). Participants were primarily from the United Kingdom (28.23%), Italy (18.72%), the United States (14.33%), and Colombia (11.06%), with the remaining 27.66% distributed across other countries. Ethical approval for this study was obtained from the University of Sussex Ethics Committee (ER/MHHE20/1).

Procedure

The study was conducted in line with the born-open principle (De Leeuw, 2024), ensuring transparency and reproducibility at every stage. The experiment was implemented with *JsPsych* (De Leeuw, 2015) and all materials, including its code, the items, instructions and exact procedure, is available in open-access at https://github.com/RealityBending/FictionEro. Note that the experiment was translated from English into French, Italian and Spanish (based on the author's native language) and then disseminated in the relevant countries.

Phase 1: Affective Evaluation. Participants were first presented with a cover story introducing the goal of the study as a validation of a new image-generation algorithm trained to produce high-quality erotic (and also non-erotic) content. Participants were informed that they would be shown a series of images (presented for 2.5 s),, some of which were generated by the algorithm and others that were real photographs. Each image would be preceded with a cue (the word "AI-Generated" or "Photo") presented for 1.25 s, and the participants' task was to rate each stimuli on several affective dimensions.

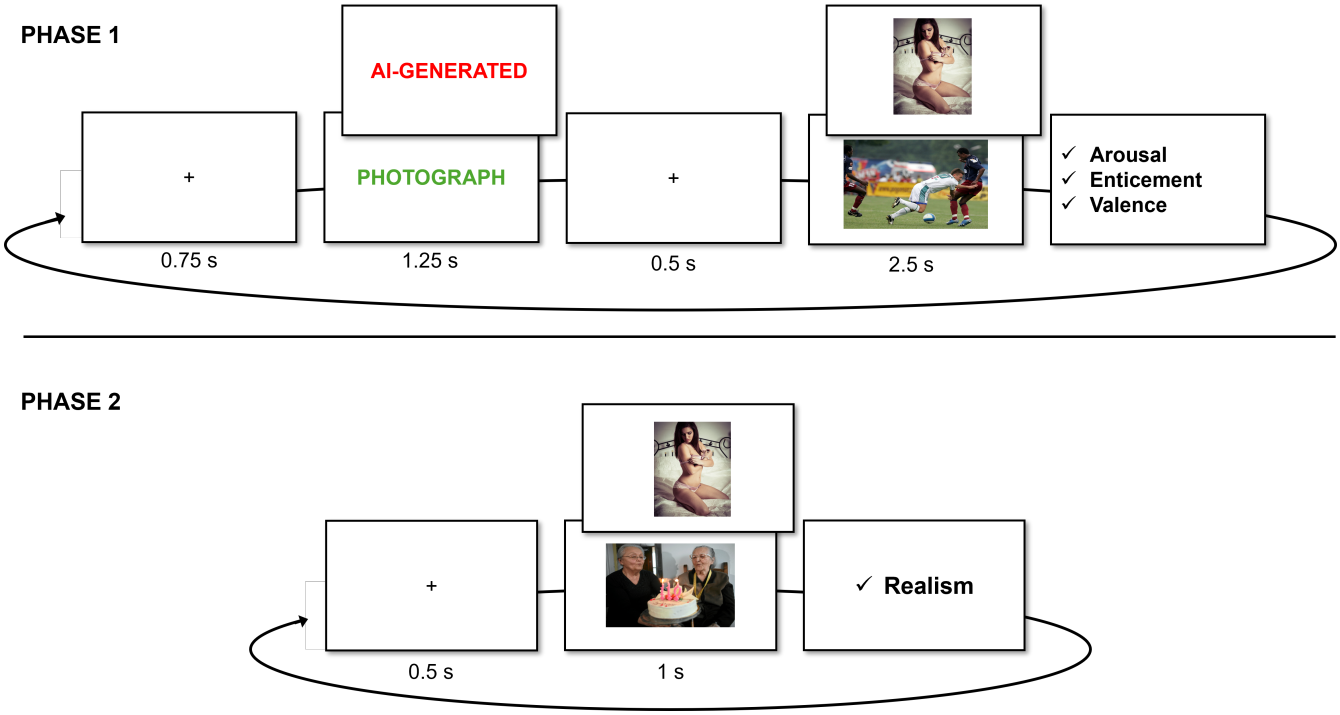
Three analog scales were presented: *Arousal* ("How much do you find the image sexually arousing? This question is about your own personal reaction felt in your body when seeing the image"), *Enticement* ("How enticing and sexually appealing would you rate this image to be. Think of how much, in general, people similar to you in terms of gender and sexual orientation would like it"), and *Valence* ("Did the image evoke a positive and pleasant (not necessarily sexual) feeling in you, or could it be better characterized as negative and unpleasant? Think of how much you did enjoy (or not) looking at the images").

Unbeknownst to participants, all 60 stimuli were real images and the cues were randomly assigned. Sixtiy stimuli were included: 40 erotic images from the NAPS ERO (Wierzba et al., 2015) and 20 non-erotic images from the original NAPS database (Marchewka et al., 2014), using the NAPS ERO normative ratings from heterosexual men and women for all images. The erotic images formed two sets of 20, one selected on men's arousal norms and one on women's. Each set comprised, within each image category (male, female), the eight highest- and two lowest-arousing images (i.e., 10 male and 10 female images per set), with the women-normed set drawn from images not already selected for the men-normed set. Non-erotic images (NAPS "Faces" and "People" categories) were selected on arousal averaged across men's and women's ratings: the 10 lowest-arousal (neutral) images, and the 10 highest-arousal images among those with mean valence above the scale midpoint (positively arousing).

Questionnaires. The affective evaluation phase was followed by a series of questionnaire. The *Beliefs about Ar-*

Figure 1

Overview of the experimental procedure for Study 1. A) Phase 1: Participants viewed a randomized sequence of erotic and non-erotic images. After each image, they provided ratings on three dimensions: sexual arousal, enticement, and affective valence. B) Phase 2: The same set of images was presented again in a new random order. This time, participants rated each image based on its perceived realism (i.e., how photographic or lifelike it appeared).



tificial Image Technology (BAIT) questionnaire (first used in Makowski, Te, et al., 2025) was refined to include, additionally to 6 items from the GAAIS (Schepman & Rodway, 2020, 2023) measuring negative and positive attitudes towards AI, 8 items specifically developed to assess beliefs about computer-generated content of various types such as text, videos, images, and VR-environments. Items from the GAAIS formed the attitudes score (ω = .72) and the rest of the items were averaged into an expectations score (ω = .70). Additionally, the first stand-alone item assesses self-reported AI knowledge on a 6-point Likert scale (Not at all - Expert).

Two items from the *Consumption of Pornography Scale* (COPS, Hatch et al., 2023) were included to assess pornography use frequency and duration. An additional item assessed the recency of any sexual activity, with response options ranging from within the past 24 hours to more than a year ago.

Phase 2: Realism Beliefs. In order to measure a proxy of the "beliefs" of participants about our manipulation and about the stimuli's nature, we devised a second phase of the experiment in which participants were asked to rate the realism of the images. The instructions were framed as follows: "In the next phase, we would like to see if you found our image generation algorithm convincing and error-free. We will briefly present you all the images one last time (the AI-generated ones, as well as the photos), and you will have to rate them on how real (how realistic, photography-like) the image is". Participants then viewed the same 60 images (1 s) in a newly randomised order and rated the stimuli on realism with an analog slider scale (AI-generated - Photograph). The expectation was that items labelled as AI-generated would be rated as less realistic than those labelled as photographs.

Following phase 2, a manipulation check was done by offering participants the opportunity to tick boxes pertaining to their experience, including items prompting whether they could distinguish which images were AI-generated or perceived no difference between photographs and AI-generated images; whether they felt the AI-generated images were more or less arousing; whether they believed the labels were inaccurate or reversed; and whether they found some images arousing or none at all. Finally, participants were debriefed and were informed of the true purpose of the study, and of the fact that all images were actually genuine pictures.

Data Analysis

Bayesian Mixed Models were fitted separately for Arousal, Valence, and Enticement, with Gender (Female

vs. Male), Condition (AI-Generated vs. Photograph), Stimuli Type (Female vs Male vs Non-Erotic Images), and Congruence (True vs. False). Congruence, used as a proxy of condition belief, was also computed based on whether the condition matched the (dichotomized) reality rating of phase 2.

As the outcome variables (measured on analog sliders) showed excess responses at the bounds, Zero–one inflated beta (ZOIB) models (Heiss, 2021; Vuorre, 2019) were used. Participants and items were entered as random intercepts, with the effects of stimuli type and condition also modeled as variable within participants. Conservative priors ($Normal(0, 0.5)$) were placed on all fixed effects. The models were run using the brms package (Bürkner, 2017) and analysed using the easystats collection of packages (Lüdecke et al., 2021, 2022; Makowski et al., 2019).

For all outcomes, we assessed the difference between the conditions (Photograph - AI-Generated) using marginal contrasts (Makowski, Ben-Shachar, et al., 2025), stratified by stimuli type, gender, and congruency. Coefficients are considered as significant when the 95% credible interval (*CI*) excludes zero. As the patterns of results for the three outcomes were similar, we will focus on presenting the detailed results for Arousal as our primary outcome, and highlighting when divergent trends are present for Enticement or Valence.

To evaluate the reliability of individual-level parameters (i.e., random intercepts and random slopes for each participant and each item), the Variance-over-Uncertainty Ratio index (*D-vour*, Lüdecke et al., 2021) will be computed. This index estimates the reliability of group-level variance by quantifying the normalised ratio of observed between-group variability to the uncertainty associated with the corresponding random-effect estimates. Values of D-vour greater than 2/3 indicate moderately reliable random-effect estimates, corresponding to a 2:1 ratio of between-group variance to estimation uncertainty. In this context, reliability refers to cases in which variability between groups (e.g., participants) exceeds the uncertainty in the estimates, suggesting the presence of meaningful inter-individual differences.

Lastly, a correlation matrix will be computed between the models' individual-level estimates and potential moderators: beliefs, attitudes towards and knowledge of AI, porn frequency and recency of sexual activity.

All analyses were conducted in R (version 4.5.2). Anonymized data, together with all preprocessing and analysis scripts, will be openly released on https://github.com/RealityBending/FictionEro/tree/main/study1/ to facilitate complete reproducibility.

Results

Manipulation Checks

An initial manipulation check was conducted to assess participants' belief in the instructional cues. Based on participants' feedback after the final rating phase, 41.7% of participants reported that they did not believe the labels, indicating that a large proportion of participants has doubts about the labels (and justifying our inclusion of the belief factor based on the participant's responses in phase 2).

To examine whether participants' explicit beliefs about AI-generated content corresponded to their affective responses, Bayesian independent-samples *t*-tests compared the difference scores between images labelled as AI-generated and as photographs as a function of whether participants reported finding AI images less arousing.

For arousal, participants who Self-reported the impression AI images were less arousing showed a larger difference between images labelled as AI-generated and as photographs (Bayesian *t*-test: BF_{10} = 3.45, median δ = 0.31, 95% *CI* [0.07, 0.54], *pd* = 99.5%), suggesting some degree of awareness. A similar pattern was observed for enticement and valence.

Effect of Stimuli Type

To ensure that different stimulus types (female, male and non-erotic images) elicited different responses, marginal contrasts from the mixed models were computed to assess differences in arousal ratings across stimulus types separately for males and females.

For males, female erotic images were rated as more arousing than both male erotic (Δ = 0.43, 95% *CI* [0.39, 0.46], *pd* = 100%) and non-erotic images (Δ = 0.44, 95% *CI* [0.40, 0.47], *pd* = 100%). For females, male erotic images were rated as more arousing than non-erotic images (Δ = 0.20, 95% *CI* [0.17, 0.24], *pd* = 100%), but not credibly more than female erotic images (Δ = 0.04, 95% *CI* [0.00, 0.08], *pd* = 96.36%). Similar patterns were found for enticement and valence; although for females, non-erotic images were rated as more positively valenced than both male erotic images (Δ = -0.08, 95% *CI* [-0.13, -0.03], *pd* = 99.80%) and female erotic images (Δ = -0.09, 95% *CI* [-0.14, -0.03], *pd* = 99.93%).

Effect of Condition

The effect of condition only emerged for males for erotic images of females, with images labelled as photographs rated as slightly more arousing than those labelled as AI-generated (Δ = 0.02, 95% *CI* [0.00, 0.03], *pd* = 99.6%). For females, weak evidence in the opposite direction was observed for erotic images of males, which were rated slightly higher in arousal when labelled as AI-generated (Δ = –0.02, 95% *CI* [–0.03, 0.00], *pd* = 98.3%).

No other contrasts provided credible evidence for differences (max. pd = 88.3%). Similar results were seen for enticement and valence ratings for males, but no credible differences emerged for females (max. pd = 92.6%)

Effect of Condition as a function of Beliefs

Estimated marginal contrasts examined the effect of condition as a function of gender, stimuli type, and congruence (see Figure 2). In congruent trials (i.e., Photograph-labelled images later rated with high realism and AI-labelled images later rated with low realism - congruent with an effective and beliefs-aligned manipulation), males rated erotic images of females labelled as photographs as more arousing than those labelled as AI-generated ($\Delta = 0.08$, 95% CI [0.06, 0.10], pd = 100%), and females showed the same pattern for male erotic images ($\Delta = 0.06$, 95% CI [0.04, 0.08], pd = 100%).

In incongruent trials (i.e., where realism ratings were not matching the experimental condition, suggesting opposite beliefs), this pattern reversed for the same stimulus types: images labelled as AI-generated were rated as more arousing than those labelled as photographs. This reversal was more pronounced in females ($\Delta = -0.09$, 95% CI [−0.12, −0.07], pd = 100%) than in males ($\Delta = -0.05$, 95% CI [−0.07, −0.03], pd = 100%). Similar crossover patterns were observed for enticement and valence ratings of erotic, relevant images.

For the remaining stimulus types (male erotic and non-erotic images for male participants; female erotic and non-erotic images for female participants), differences between the label conditions were generally smaller and less clear. In congruent trials, males rated non-erotic images labelled as AI-generated as slightly less arousing than those labelled as photographs ($\Delta = 0.02$, 95% CI [0.01, 0.04], pd = 99.66%), and a comparable effect was observed in females for female erotic images ($\Delta = 0.02$, 95% CI [0.01, 0.04], pd = 100%). In incongruent trials, this pattern reversed; but a credible difference emerged only for females' ratings of female erotic images, which were higher when labelled as AI-generated ($\Delta = -0.03$, 95% CI [−0.05, −0.01], pd = 99.69%). For enticement ratings the same reversal pattern was found, with credible differences for all with the exception of males' ratings of non-erotic images. On the other hand, all differences were credible for valence ratings for both genders and the same crossover effect was found.

Individual-level parameters and correlations with moderators

While item- and participant-level intercepts showed high reliability (D-vours ≥ 0.90), very low reliability was found for participant-level interactions with the AI-generated condition across all stimulus type parameters (D-vours = 0.008–0.135), indicating minimal reliable inter-individual variability in responses to the labelling manipulation. Broadly similar patterns emerged for enticement and valence, except that participant-level variability for the remaining stimulus types (i.e., irrelevant and non-erotic images) was reliable for these outcomes (D-vours ≥0.835).

Participant-specific intercepts, stimulus type, and AI-generated parameters showed only weak correlations (all $|rs| \leq 0.12$) with sexual behaviours and with attitudes and expectations towards AI, indicating that individual variability in these parameters was not meaningfully related to these constructs. The same pattern emerged for enticement and valence (full analysis available at the link above).

Discussion

In the first study, we presented participants (heterosexual males and females) with images from the NAPS database, including non-erotic images and erotic images of both males and females, each presented as either "photograph" or "AI-generated". After collecting subjective ratings of arousal, enticement and valence for each image, we assessed the efficacy of the labelling manipulation by asking participants to rate how realistic they deemed each image.

As per H2, marginal contrasts indicated that males rated erotic images of females (i.e., the category corresponding to their sexual interest) as more arousing than erotic images of males, whereas females did not credibly differentiate between erotic images of males and females. This is consistent with prior literature demonstrating stronger category-specificity in men's subjective responses to explicit stimuli (Chivers et al., 2004; Chivers, 2017; Huberman et al., 2015). Prior work also suggests that women's sexual responses are more sensitive to contextual and relational features, such as narrative, emotional connection, and cues of intimacy (Laan et al., 1994; Rupp & Wallen, 2007, 2009).

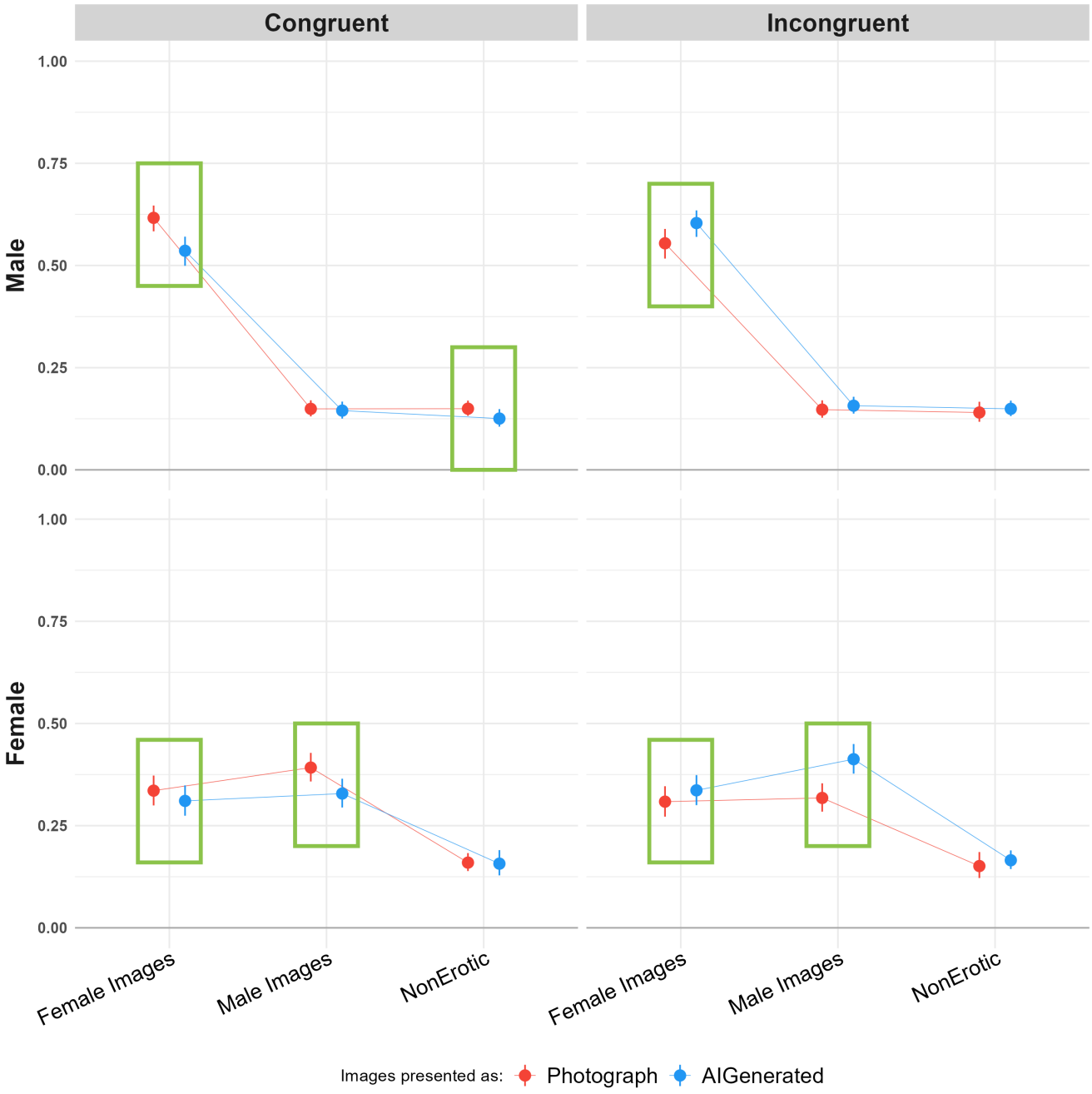
In alignment with our H1 and with Marini et al. (2024), we observed an anti-AI bias effect that only clearly emerged for males' arousal ratings of erotic images of females (i.e., the stimulus type yielding the highest arousal ratings). While this effect was seemingly absent for women, the pattern of results became clearer once taking participants' own elaboration and beliefs into account (H3), namely the congruency between the label and participants' beliefs about the image (as indexed by our realism measure in phase 2).

For items for which the label and the realism rating were congruent, the expected pattern emerged: images labelled as AI-generated were rated as less arousing than those labelled as photographs, for both male and female participants. But for items for which the label and the realism rating were incongruent, this pattern reversed, suggesting an effect driven more by participants' beliefs about the image than by the label itself.

We also tested whether variability in responses could be explained by individual differences in participants' attitudes towards AI (H4) and their sexual behaviour (H5). To do so, we first assessed the extent to which responses to the

Figure 2

Estimated means for the effect of Condition depending on Congruency. Green rectangles represent differences in means where the pd > 97.5%, indicating a consistent effect direction (Makowski et al., 2019). A crossover effect can be seen where the difference between arousal ratings between images labelled as AI-generated and those labelled as photographs is reversed for incongruent trials. Of note, female images are relevant for heterosexual males while irrelevant for heterosexual females.



AI label varied meaningfully between participants. Between-participant variability in sensitivity to the AI label was minimal, and correlations between individual-level estimates and participants' attitudes or sexual behaviour were accordingly weak and uninformative.

Beyond these findings, Study 1 surfaced several limitations. First, participant feedback indicated that the manipulation was not fully convincing (41.7% reported disbelieving the labels), which together with the small size of the labelling effect and the minimal between-participant variability in label sensitivity, suggests that the manipulation may not have been sufficiently strong or salient, and that the cover story could be strengthened. Second, the realism ratings of Phase 2 are an imperfect proxy for participants' beliefs about image origin, as it is arguably possible to find an image highly realistic while still believing it to be AI-generated. Third, feedback suggested that the non-erotic images were experienced as distracting. Moreover, presenting identical material to all participants meant that male participants viewed erotic images of males, which they rated as strongly negative, introducing a type of valenced experience with no equivalent for female participants and complicating interpretation. Finally, the stimuli did not include depictions of couples, as multi-person images were notably difficult to generate realistically at the time, and their inclusion could have made the cover story suspicious. Given women's greater sensitivity to relational cues, this absence of relational context may have contributed to the lower arousal ratings among female participants.

Study 2

The goal of Study 2 was to replicate the main findings of Study 1 with a refined paradigm addressing the limitations identified above.

Regarding the stimuli, as the limitations in genAI realism that had prevented the inclusion of couple images had largely faded by the time of Study 2, couple-based stimuli were included to assess whether the absence of relational context contributed to the lower arousal ratings among female participants. Additionally, non-erotic images were removed, and erotic stimuli depicting individuals were tailored to participants' gender (images of females for male participants and vice versa), alongside a fixed set of couple images presented to all participants.

Regarding the manipulation, Study 2 employed a more elaborate cover story to strengthen its credibility, framing the task as a collaboration with a (fictional) AI startup and including visual details about how AI generation works. Finally, to obtain a more direct measure of beliefs, Phase 2 was reframed as a "pseudo-reveal": participants were told that, while the images were indeed a mix of photographs and AI-generated images, the labels had been assigned at random, and were asked to identify which images they believed to be AI-generated. This reframing allowed for a more direct measure of beliefs without compromising the true nature of the experiment (the fact that all images were actually real photographs).

Based on these aims, we tested the following hypotheses. As in Study 1, we hypothesised that images labelled as AI-generated would be rated lower in arousal than those labelled as photographs (H1). We also hypothesised that the inclusion of couple-based stimuli would narrow the gender gap in arousal observed in Study 1 (H2). Given the interaction effect with Realism ratings in Study 1, we hypothesised that this pattern would be even stronger, based on our improved measure of beliefs (H3). Finally, as in the previous study, we tested whether attitudes towards AI (H4) and sexual activity and pornographic consumption (H5) moderate responses to the stimuli.

Materials and Methods

Participants

The initial sample comprised 279 participants recruited via Prolific©. Inclusion criteria required participants to be native English speakers or residents of countries with high levels of English proficiency. Data collection occurred from 27 of January 2025 to 31 of July 2025. By this time, publicly available image generators were producing outputs that people could no longer reliably distinguish from authentic photographs (Cooke et al., 2025).

Participant exclusions were applied as follows: five participants were removed for showing no variability in arousal ratings (i.e., they did not move the response scales). An additional five participants were excluded for completing the study on a mobile device. One participant was excluded due to displaying negative correlations between arousal and both enticement and valence. Furthermore, five participants who self-identified as neither female nor male, and two participants who reported a sexual orientation other than heterosexual, homosexual, or bisexual, were excluded from further analyses due to lack of sufficient group sizes. Finally, one participant was removed because the stimuli presented were not relevant to their gender and sexual orientation.

The final sample consisted of 261 heterosexual participants (Mean age = 37.4 ± 12.7, 48.7% Female). 56.32% of participants were from the United Kingdom, 26.82% from South Africa, 10.34% from the United States, and the remaining 6.51% were from other countries.

Ethical approval for this study was obtained from the School of Psychology Ethics Committee at the University of Sussex (ER/EB672/2).

Procedure

Phase 1: Affective Evaluation. In this study, participants were informed that the researchers were collaborating

with a young AI start-up based in Brighton, UK, specializing in making more ethical AI content. This was intended to enhance the believability of the study. Participants were then informed that they would be shown a series of highly realistic AI-generated erotic images intermixed with real photographs, each preceded by a cue (as in Study 1) presented for 1s.

The *Arousal*, *Enticement* and *Valence* ratings were defined as per Study 1 (see above) but this time they were rated on a 7-point Likert scale from Not at all (0) to Very much (6).

As in Study 1, all stimuli were real photographs (presented for 2 s) from the NAPS ERO (Wierzba et al., 2015) and the labels were randomly assigned. Unlike Study 1, no non-erotic images were included. Each participant viewed 50 images relevant to their gender and sexual orientation: 25 depicting individuals and 25 depicting opposite-sex couples. For the female-individual and male-individual categories, all 25 available NAPS ERO images were used. For opposite-sex couples, we selected the 25 (of 50 available) with the highest arousal summed across heterosexual men's and women's normative ratings; among three images tied at the selection threshold, the one with the lowest summed valence was excluded.

After viewing all images, participants completed a brief feedback asking them to indicate whether they found some or no images particularly arousing, and whether AI-generated images were more or less arousing than the photographs. Participants were also asked whether differences between AI-generated and real images were obvious or subtle, whether they perceived any inconsistencies or reversals in the labeling, and whether they believed all images were either photos or AI-generated. If participants indicated that they believed all images were either real or AI-generated, they were further asked to rate their confidence in this judgment on a scale from "Not at all" to "Completely certain".

Questionnaires. The questionnaires used in Study 2 were largely the same as those in Study 1, with minor modifications. In the BAIT scale, two items, assessing beliefs that AI might take control of people and interest in using AI systems in daily life, were removed. Three items assessing AI expertise were added (e.g., "I consider myself an expert in AI technology"), as well as four items assessing one's ability to discriminate between AI and real stimuli (e.g., "I can easily distinguish between real and AI-generated images"). Lastly, 7 items assessing AI-bias were further included (e.g., "Human creators bring a unique perspective that AI cannot replicate"). This version of the BAIT resulted in 5 dimensions: Expectations ($\omega = .76$), Attitudes ($\omega = .78$), Expertise ($\omega = .77$), Discrimination ($\omega = .80$) and Bias ($\omega = .77$). Items were rated in a 7 Likert scale from 0 (Disagree) to 6 (Agree). Additionally, the wording was streamlined by replacing "Artificial Intelligence" with "AI" throughout the scale.

In the COPS, the item assessing the typical duration of pornography viewing sessions was omitted, retaining only items measuring frequency of pornography viewing and recency of sexual activity.

Phase 2: Reality Beliefs. Phase 2 replicated that of Study 1, presenting participants with all images again (for 1 s) in a randomised order and assessing the perceived authenticity of the stimuli. The instructions, however, made the manipulation more overt by explicitly disclosing the mislabeling: "But there is something important we need to tell you… In the previous phase, some images were intentionally mislabeled (we told you it was a photograph when it was actually AI-generated, and vice versa)." Participants were then asked to judge the true category of each image by indicating their confidence (i.e., "Indicate your confidence that the image is fake or real") on a continuous scale from AI-Generated (−3) to Photograph (3).

Following completion of phase 2, participants provided general feedback regarding their experience and any additional comments. Finally, participants were presented with a debrief page informing them that all images were, in fact, real photographs.

Data Analysis

As in Study 1, Bayesian linear mixed models will be fitted for the three outcome variables, each predicted from Gender (Female vs. Male), Condition (AI-generated vs. Photograph), and Congruence (True vs. False). Congruence, again used as a proxy of condition belief, was computed based on whether the condition matched the (dichotomized at the midpoint) reality judgement of Phase 2. For ease of comparison with study 1 results, all outcome variables will be bounded between 0 to 1, and the ZOIB family will be fitted. Conservative priors ($Normal(0, 0.5)$) will be placed again on all fixed effects.

Marginal contrasts will be computed to compare marginal means with condition (i.e., AI-generated and Photograph) as the predicted contrasts, stratified by gender and congruency. The same individual-level parameter and moderator-correlation analyses as in Study 1 will be conducted, using the same packages. To further assess the effect of the type of image (e.g., couple vs individuals) the same analysis will be conducted adding the type of image (image of individuals vs image of couples) as predictor.

Consistent with Study 1, the Results section will focus on Arousal as the primary outcome, highlighting where divergent trends emerge for enticement or valence; full results for all outcomes are available in the analysis scripts on GitHub. All analyses were conducted in R (version 4.5.2), and all materials, data, and analysis scripts for Study 2 are available at https://github.com/RealityBending/FictionEro/tree/main/study2.

Results

Manipulation checks

In Study 2, 32% of participants reported believing that all images were real based on their feedback after Phase 1, indicating that a third of participants did not believe in the manipulation.

To examine whether participants' explicit beliefs about AI-generated content corresponded to their affective responses, Bayesian independent-samples t-tests compared the difference scores between images labelled as AI-generated and as photographs as a function of whether participants reported finding AI images less arousing. For arousal, participants who reported finding AI images less arousing showed a numerically larger difference between images labelled as AI-generated and as photographs ($M_{difference}$ = −0.03) than those who did not ($M_{difference}$ = −0.009), but there was no credible evidence for a group difference (BF_{10} = 0.47; median δ = 0.24, 95% *CI* [−0.18, 0.66], *pd* = 88.4%). Similarly, no credible group differences emerged for enticement or valence.

Effect of Stimuli Type

The addition of image type (i.e., images of couples vs. images of individuals) revealed a credible gender difference in ratings of individual images, with males rating them higher in arousal than females (Δ = 0.17, 95% *CI* [0.11, 0.22], pd = 100%). No credible differences between males and females were found for ratings of couple images (Δ = 0.009, 95% *CI* [-0.02, 0.04], pd = 69.92%). The same patterns were found for enticement and valence (see Figure 3).

Additionally, credible differences between image types were found within each gender. Females rated couple images higher in arousal than images of individuals (Δ = 0.08, 95% *CI* [0.03, 0.12], *pd* = 99.98%), whereas males rated couple images lower (Δ = -0.08, 95% *CI* [-0.12, -0.04], *pd* = 99.98%). Enticement and valence ratings were also credibly different for both genders (pd > 99.99%), with males rating couple images lower than images of individuals, and females showing the reversed pattern.

Effect of Condition (with and without Type)

Estimated marginal contrasts assessing the effect of condition across gender showed that the difference was only credible for Males (Δ = 0.02, 95% *CI* [0.01, 0.03], *pd* = 99.84%) but not for females (Δ = 0.01, 95% *CI* [0.00, 0.02], *pd* = 93.57%). The same pattern was found for enticement. For valence ratings, both genders rated images labelled as photograph higher than those labelled AI-generated (pd > 98.8%).

The effect of condition across gender for the type of images showed only males credibly differed in arousal ratings of images of couples labelled photograph vs those labelled AI-generated (Δ = 0.02, 95% *CI* [0.00, 0.04], *pd* = 99.23%), and ratings of individuals (Δ = 0.02, 95% *CI* [0.00, 0.03], *pd* = 98.77%). No credible differences were found for females (pd < 88.2%). Enticement ratings followed the same pattern, while for valence ratings the only credible difference was found for females ratings of couple images, with those images labelled photograph being rated higher than those labelled AI-generated (Δ = 0.02, 95% *CI* [0.01, 0.04], *pd* = 99.52%).

Effect of Condition Depending on Congruence (with and without Type)

Estimated marginal contrasts examined differences in arousal ratings between images labeled as photographs or AI-generated as a function of gender and congruence (see Figure 4). Under congruent trials (i.e., where Phase 2 reality judgements matched the label, suggesting the labels were believed), images labelled as photographs were rated as more arousing for both genders, with a stronger effect for males (Δ = 0.06, 95% *CI* [0.04, 0.07], pd = 100%) than females (Δ = 0.02, 95% *CI* [0.00, 0.04], *pd* = 97.18%). Under incongruent trials (i.e., where judgements opposed the label, suggesting disbelief), the direction of the effect reversed for males (Δ = −0.02, 95% *CI* [−0.04, −0.01], *pd* = 99.71%), whereas the effect for females was negligible (Δ ≈ 0, 95% *CI* [−0.02, 0.02], *pd* = 57%).

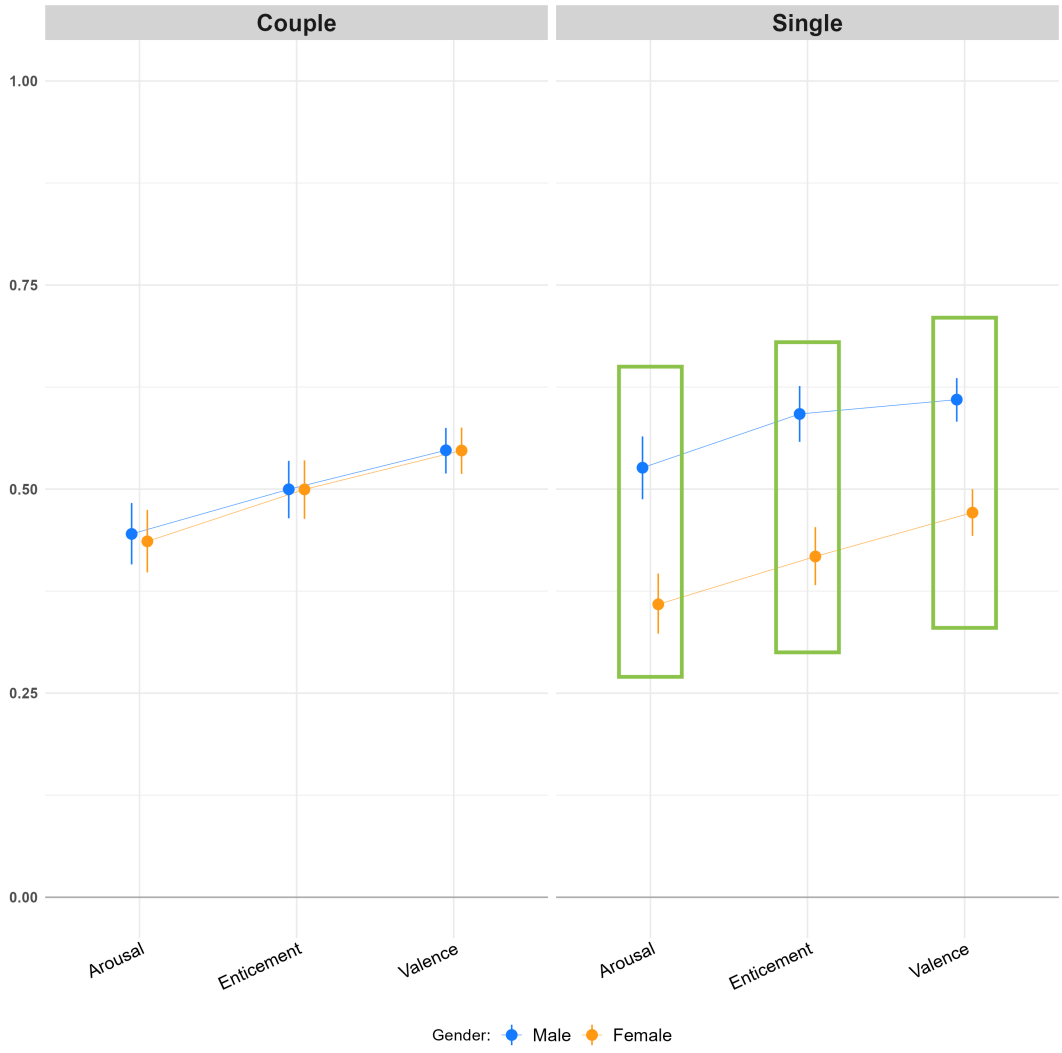
Incorporating image type into the models provided limited additional evidence for differences overall, although some credible effects emerged for males. Specifically, under congruent trials, males showed higher arousal ratings for photograph- relative to AI-labelled images for both couples (Δ = 0.06, 95% *CI* [0.03, 0.08], *pd* = 100%) and individuals (Δ = 0.06, 95% *CI* [0.04, 0.08], *pd* = 100%). Under incongruent trials, a credible difference for males was observed only for individual images (Δ = −0.03, 95% *CI* [−0.05, −0.01], *pd* = 99.41%). In contrast, female participants showed no credible differences across trial types or image categories, with estimates centred near zero and credibility intervals spanning zero.

Enticement and valence ratings followed a broadly similar pattern to arousal, with higher ratings for photographic images under congruent trials and weaker, less consistent reversals under incongruent trials. A notable exception was observed for female participants, who showed credible differences under congruent trials, with photograph-labelled images rated higher for both enticement and valence (pd > 99.2%).

When image type was included in the models, credible differences in enticement ratings were largely restricted to males: under congruent trials, males showed higher ratings for photograph-labelled images for both couples and individual stimuli (pd = 100%), whereas under incongruent trials, this effect remained only for individual images (pd > 99%). For valence ratings, most contrasts became credible follow-

Figure 3

Estimated means for Arousal, Enticement and Valence for images of individuals and couples, for both genders. Green rectangles represent significant differences.



ing the inclusion of image type (pd > 98%), maintaining the same reversal pattern under incongruent trials. The primary exception was for female ratings of couple images under incongruent trials, where evidence remained inconclusive (pd < 59.5%).

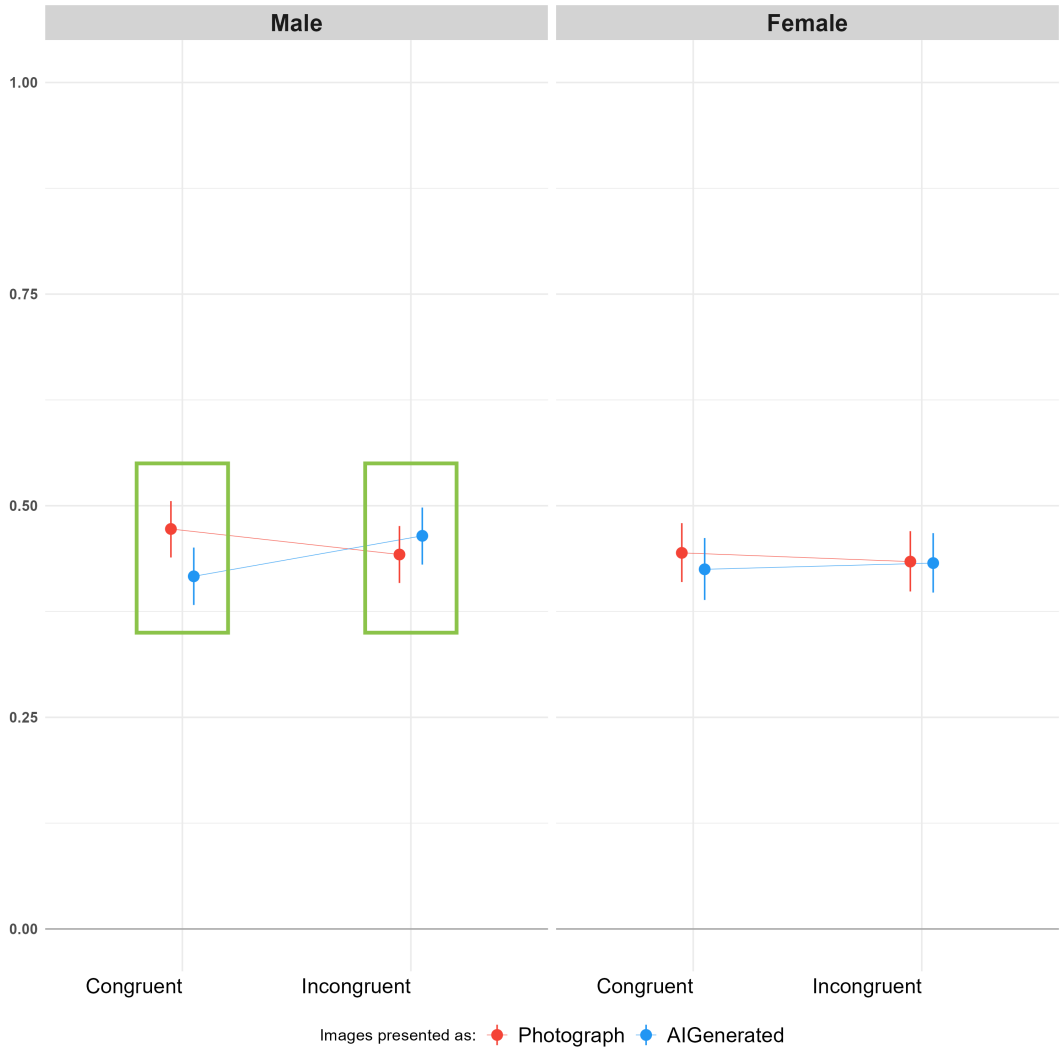
Individual-level parameters and Correlations with moderators

While participant- and item-level parameters showed high reliability (D-vours > 0.91), very low reliability was found for participant-level interactions with the AI-generated condition (D-vour = 0.072), indicating minimal reliable inter-individual variability in responses to the belief manipulation. A similar pattern emerged for enticement and valence, although the interaction showed slightly higher (but still low) reliability for valence (D-vour = 0.203).

Participant-specific intercepts and AI-generated parameters showed only weak correlations with AI-related bias, expectations, and attitudes, discrimination ability, and sexual behaviours (all $|rs| \leq 0.16$), with the strongest associations observed for overall BAIT scores (r = 0.23, 95% *CI* [0.22, 0.25]) and self-reported AI expertise (r = 0.25, 95% *CI* [0.23, 0.27]). Similar pattern for enticement and valence were found (full analysis available at the link above). Taken together these findings suggest that individual variability in these parameters was not meaningfully related to these constructs.

Figure 4

Estimated arousal means for the effect of Condition depending on Congruency. Green rectangles represent significant differences.



Discussion

In Study 2, participants viewed erotic images depicting both couples and individuals matched to their sexual orientation. As in Study 1, images were labelled as either photographs or AI-generated. Following Phase 1, 32% of participants reported believing that all images were real. Although this measure differs from Study 1's manipulation check (where 41.7% reported disbelieving the labels in any form) and the two figures are therefore not directly comparable, the pattern is at least consistent with the strengthened cover story in Study 2 having improved the manipulation's credibility.

Overall, the pattern of findings partially replicated those observed in Study 1. Considered independently of congruence, the AI-labelling effect on arousal was credible for males but not for females, providing support for H1 in males only.

A key aim of Study 2 was to examine whether the inclusion of relational context (i.e., via couple-based stimuli) would narrow the gender gap in arousal observed in Study 1 (H2). The findings partially support this prediction. Specifically, males and females did not differ credibly in their ratings of couple images, whereas males continued to rate individual images as more arousing than females. However, this convergence did not arise solely from the predicted mechanism of selectively increased female arousal: females rated couple images higher in arousal than images of individuals, but males rated couple images credibly lower than images

of individuals. The gender gap therefore appears to have narrowed through change in both directions (i.e., increased arousal to couples among females and decreased arousal to couples among males) rather than through a female-specific increase.

As per H3, under congruent trials, images labelled as photographs were rated as more arousing, enticing, and positively valenced compared to those labelled as AI-generated. Under incongruent trials, this pattern reversed for males, consistent with the crossover effect observed in Study 1. For females, however, no credible reversal emerged under incongruent trials, indicating an absence of the predicted effect rather than merely a weaker version of it; some credible differences did emerge for females under congruent trials for enticement and valence ratings. Thus, while the predicted crossover interaction was observed, support was essentially confined to male participants, only partially replicating Study 1.

The overall effect of AI labelling remained small and showed minimal reliable variability across participants, replicating the pattern observed in Study 1. Consequently, as in Study 1, it was not possible to draw strong conclusions regarding individual differences as moderators of the labelling effect (H4-H5). Weak associations were found between arousal responses (both baseline and AI-related parameters) and self-reported beliefs, attitudes, and AI expertise (see full analysis on the link provided above).

General Discussion

The present paper investigated the anti-AI bias in response to erotic stimuli across two studies. Overall, the findings provide converging evidence for three main conclusions: (1) there is a consistent but small effect of AI labelling on responses that is primarily observed in males, (2) there is a strong gender difference in response to erotic stimuli, that is robust and context dependent, and (3) belief-related processes (i.e., congruency between initial labels and subsequent realism/reality judgements) play a central role in shaping affective responses.

In line with previous findings (e.g., Marini et al., 2024), images labelled as AI-generated were generally rated as less arousing than those labelled as photographs. However, across both studies this effect was small in magnitude and primarily driven by male participants, whose ratings were higher in arousal, enticement, and valence, for images labeled as photographs than for those labeled as AI-generated. On the contrary, in Study 1, females rated relevant erotic images higher in arousal (but not valence or enticement) when they were presented as AI-generated. This reverse pattern was not found in Study 2.

Our results also replicate and extend prior literature on gender differences in response to erotic stimuli (Chivers et al., 2004; Chivers, 2017; Huberman et al., 2015; Safron et al., 2020). In Study 1, males rated sexually relevant images as more arousing than irrelevant ones, whereas females did not. Study 2, which presented only relevant stimuli, examined whether the gender gap in arousal observed in Study 1 was contingent on the nature of the stimuli. When relational context was introduced through couple-based images, this gap narrowed substantially: males and females did not differ in their ratings of couple images, while a gap persisted for images depicting individuals. Notably, Study 2 also showed that females rated images of couples higher than images of individuals, while males did the opposite. This finding supports the interpretation that female sexual responses are more sensitive to contextual and relational cues (Laan et al., 1994; Rupp & Wallen, 2007, 2009) and suggests that the absence of such features in Study 1 likely amplified the apparent gender gap in affective reactions.

Third, and most importantly, the present findings highlight the central role of belief-related processes in shaping affective responses. In both studies, when asked for feedback, a substantial proportion of participants expressed doubts about the veracity of the labels. We further evaluated these belief-related effects on ratings by operationalising a congruency variable, which revealed a crossover interaction with condition. When participants' realism judgments were congruent with the initial label, images labelled as photographs were rated higher than those labelled as AI-generated. In contrast, when judgments were incongruent, this pattern reversed, with AI-labelled images receiving higher ratings, suggesting that participants' affective responses are closely tied to whether they were second-guessing the authenticity of the images.

We had additionally hypothesised that individual differences in AI attitudes and beliefs, pornography consumption frequency and recency of sexual activity might be associated with the strength of this labelling effect. However, individual-level estimates of sensitivity to the AI-generated label showed minimal reliable variability between participants, meaning these hypotheses could not be meaningfully tested in the present data; weak correlations with these variables should accordingly not be read as evidence against an association, but as uninformative given the lack of reliable variance to explain.

Our findings suggest that anti-AI bias can extend to highly salient erotic stimuli, but only insofar as AI attribution is believed. Thus, the anti-AI bias is better understood as a belief-sensitive authenticity effect than as a uniform response to AI labels.

This result has relevant implications for research using synthetic stimuli. Attracted by the clear advantages in terms of controllability, scalability, and stimulus standardization, several researchers are promoting the use of AI-generated stimuli in psychological research (Albohn et al., 2022; Becker et al., 2024; Becker & Laycock, 2023; Chiu

et al., 2026). However, their perceived origin must be controlled. Researchers using AI-generated erotic, affective, or social stimuli should therefore measure, manipulate, or report participants' beliefs about stimulus authenticity, rather than assuming that visual realism alone makes synthetic and photographic stimuli psychologically interchangeable.

Our results also corroborate the hypothesis that the 'authenticity aura' might indeed enhance the appeal of sexual content perceived to be 'genuine' over AI-generated content (Easterbrook-Smith, 2025). This might yield ambivalent effects. On the one hand, the general skepticism about content authenticity in the face of the rise of AI-generated content might dampen the incentives to spread non-consensual sexual material, as proposed by Viola and Voto (2023). On the other hand, this reduced affective reaction might translate into a 'moral discounting' and actually promote the spread of such content, based on the (false) assumption that explicitly fake non-consensual sexual deepfakes are not harmful (see Flynn et al., 2022).

A final consideration pertains to the possible impact of disclosing deepfakes of sensual (though not necessarily sexual) pictures depicting human bodies in a wider societal context. Recall that some legislation, notably the recent EU AI Act (Art. 50), will soon mandate the explicit disclosure of the AI-generated origin of some media. An unexpected effect of such legislation when it comes, for instance, to advertising perfume or swimwear, might be that AI-generated models, while explicitly flagged as such, could lose some of their allure with respect to their human counterparts.

Interpreting the results of this study requires particular caution due to the nuance of its design. As expressed above, it is worth recalling that a substantial proportion of participants expressed doubts about the manipulation (in Study 1, 41.7% reported disbelieving the labels; in Study 2, 32% believed all images were real), meaning the experimental manipulation was not fully convincing for a meaningful portion of the sample. We contend, however, that this limitation extends to all similar studies, stressing the relevance of including explicit attention checks before drawing any conclusion.

Additionally, while the present findings are consistent with a crossover effect linking arousal and perceived authenticity, the design cannot establish the direction of this relationship. The observed congruency effects are equally consistent with authenticity impressions shaping subsequent arousal, or with arousal being retrospectively used to infer authenticity. Resolving this ambiguity would require designs capable of manipulating, rather than only measuring, one of these two processes. Moreover, future studies are needed to establish whether these effects extend to non-heterosexual samples.

To summarise, we investigated the anti-AI bias for erotic stimuli by conceptually replicating Marini et al. (2024). Participants viewed erotic (Studies 1 and 2) and non-erotic (Study 1) images, each preceded by a randomised cue indicating that the image was either a photograph or AI-generated. Across both studies, AI labelling produced consistent but small effects, driven primarily by male participants, alongside robust gender differences in responses to erotic stimuli that proved sensitive to relational context. Most importantly, affective responses depended on whether participants believed the labels: when realism judgements were congruent with the assigned label, images labelled as photographs were rated as more arousing than those labelled as AI-generated, whereas this pattern reversed when the labels were disbelieved. Individual differences in AI attitudes, pornography consumption, and sexual activity could not account for the labelling effect, as sensitivity to the AI label showed little reliable variability across participants. Together, these findings suggest that the anti-AI bias is best understood not as a uniform response to AI labels but as a belief-sensitive authenticity effect, one with practical implications for researchers using synthetic stimuli and for policies mandating the disclosure of AI-generated content.

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